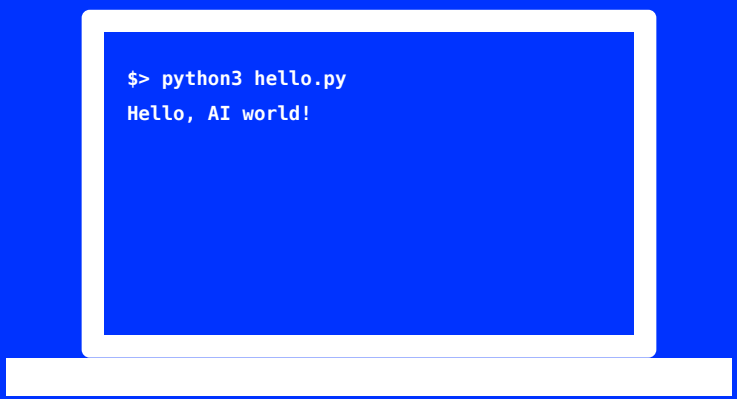
{ PISCINE IA }

DAY 01

< PYTHON ENVIRONMENT />

Week 1 - Python for AI



v 1.0

DAY 01

First things first - Rules of the piscine

Re-read the 7 rules of the piscine: never copy code you do not understand, one exercise per concept, one concrete result per week, explain without notes, document on GitHub, read the official docs, accept mistakes.

Activity	Time	Today
Course and understanding	40 min	What is a model? Python setup
Practical exercises	45 min	Tasks 01 to 03
Programming	60 min	Tasks 04 and onwards
AI project (Legal-Benin)	30 min	Legal angle of the day
Review and documentation	15 min	Self-check + push

> **i Tip:** Today is about **setting up your lab** and **understanding what an AI model really is**. Take your time: a clean environment saves you hours later.

Task 00 - Your lab: Git and GitHub

All your work will live in one GitHub repository. Create a **private** repository named **piscine-ia**, then connect your terminal to GitHub with an SSH key.

- ✓ create (or reuse) your GitHub account,
- ✓ generate an SSH key and add the public key to GitHub (Settings > SSH keys),
- ✓ check the connection and configure your identity,
- ✓ clone the repository and create the folder `week01/day01/`.

Terminal

```
$> ssh-keygen -t ed25519 -C "you@example.com"
$> cat ~/.ssh/id_ed25519.pub
$> ssh -T git@github.com
Hi <username>! You've successfully authenticated...
$> git config --global user.name "Firstname Lastname"
$> git config --global user.email "you@example.com"
$> git clone git@github.com:<username>/piscine-ia.git
```

> **i Tip:** Never share your **private** key (the file without `.pub`). Only the `.pub` file goes to GitHub.

Delivery: a first commit containing a `README.md` that states your goal for the 16 weeks.

Task 01 - Python environment

Install Python 3.10 or newer and create an isolated **virtual environment** for the piscine. A virtual environment keeps the libraries of this project separate from the rest of your computer.

Terminal

```
$> python3 --version
Python 3.11.9
$> python3 -m venv .venv
$> source .venv/bin/activate
(.venv) $> pip install --upgrade pip numpy
(.venv) $> pip freeze > requirements.txt
```

Delivery: `week01/day01/task01/requirements.txt` and a `.gitignore` at the root of the repository that ignores at least `.venv/` and `__pycache__/`.

⚠ **Warning:** On Windows, activate with `.venv\Scripts\activate`. If you use WSL, work inside the Linux filesystem, not in `/mnt/c`.

Task 02 - Hello, AI world

Write `task02/hello.py`. It must print the line below, nothing else.

Terminal

```
$> python3 hello.py
Hello, AI world!
```

i Tip: Open the Python interpreter by typing `python3` alone. It is called the REPL: test one-line ideas there, then put the final code in a file.

Task 03 - sysinfo

Write `task03/sysinfo.py`, which prints three lines: the Python version, the operating system name, and the current working directory. Use only the standard library (`sys`, `platform`, `os`).

Terminal

```
$> python3 sysinfo.py
Python: 3.11.9
System: Linux
Directory: /home/you/piscine-ia/week01/day01/task03
```

i Tip: Read the official documentation of the `platform` module. Rule 6: the docs come before any tutorial.

Task 04 - Google Colab

Colab gives you free GPUs, which you will need from week 4. Open a new notebook and run two cells:

- ✓ a cell that prints your Python version with `sys.version`,
- ✓ after switching Runtime > Change runtime type > **T4 GPU**, a cell with `!nvidia-smi`.

Delivery: `task04/day01_colab.ipynb` (File > Download > .ipynb), with the outputs visible.

> ⚠ **Warning:** A Colab session is temporary: files disappear when it stops. Always download or save to GitHub what you want to keep.

Task 05 - What is a model, really?

Write `task05/answers.md` with your own words (3 to 5 sentences each):

1. What is the difference between an AI and an AI model?
2. What is a neural network?
3. How does an AI learn from data?
4. What is a dataset?
5. What is the difference between training a model and using an already trained model?

Then add a section **Cat detector**: explain step by step how you would teach a computer to recognise a cat from 1,000 images (what data, which labels, how you check that it works, what happens with a photo it has never seen).

> ⚠ **Warning:** First draft by yourself, **no AI assistant**. Afterwards you may ask an assistant to correct you, and note what you got wrong in a `Corrections` section. That is how you learn.

Task 06 - push.sh

Write a bash script `push.sh` at the root of the repository. It takes a commit message as parameter, adds all files, commits, pushes, and prints a clear error if the parameter is missing or if the push fails.

Terminal

```
$> ./push.sh "day01: environment ready"
$> ./push.sh
Usage: ./push.sh "commit message"
```

> **i Tip:** Do not forget the shebang (`#!/usr/bin/env bash`) and `chmod +x`. Retrieve parameters with `$1` and `$#`.

> ⚠ **Warning:** Use this script only on your own repository. On a shared repository, review before you push.

Self-check (no notes!)

Rule 4 of the piscine: you must be able to explain your work without looking at your notes. Answer these out loud, then write one sentence each in **SELFCHECK.md**.

1. What is the difference between a virtual environment and your global Python?

2. Why do we ignore `.venv/` in Git?
3. In your own words: what does a model contain after training?

Delivery

Everything goes in your repository, in **week01/day01/**, with one commit per task (clear messages, in English). Run **python3 -m py_compile** on each script before pushing.

> ⚠ **Warning:** Never push your virtual environment (`.venv/`), notebooks with secrets, or huge data files. A `.gitignore` is part of your delivery.